Section 3.2 Product & Quotient Rule

Ex ① $y = (x+7)^8(x+11)^{10}$

$$[A \cdot B]' = A' \cdot B + A \cdot B'$$

$$y' = 8(x+7)^7 \cdot (x+11)^{10} + (x+7)^8 \cdot 10(x+11)^9$$

$$y' = (x+7)^7(x+11)^9 \cdot [8(x+11) + 10(x+7)]$$

$$y' = (x+7)^7(x+11)^9(18x+158)$$

$$\left[\frac{A}{B}\right]' = \frac{A' \cdot B - A \cdot B'}{B^2}$$

Ex ② $f(x) = \dfrac{3x-7}{2x+8}$

$$f'(x) = \frac{3(2x+8) - (3x-7)2}{(2x+8)^2}$$

$$f'(x) = \frac{6x+24 - 6x + 14}{(2x+8)^2}$$

$$f'(x) = \frac{38}{(2x+8)^2}$$

$$\frac{d}{dx}\left(e^x\right) = e^x$$

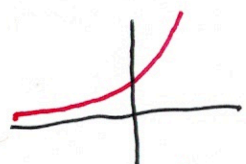
<u>Ex</u> ③ $y = x^2 \cdot e^x$ where is $y' = 0$?

$$y' = 2x \cdot e^x + x^2 \cdot e^x$$

$$0 = x \cdot e^x (2 + x)$$

$\boxed{x = 0}$ or $e^x = 0$ or $2 + x = 0$

$$\boxed{x = -2}$$



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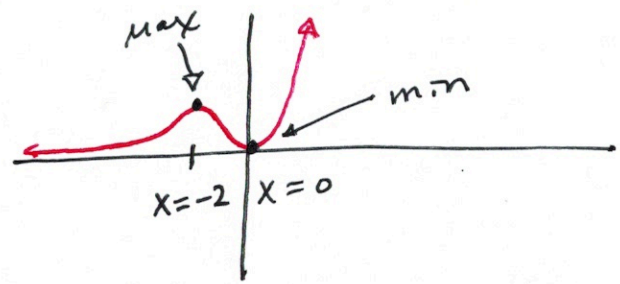

Ex (4) $f(x) = \dfrac{e^x}{x+1}$ Find the eq. of tangent line when $x = 0$.

$$y = \frac{e^0}{0+1} = 1$$

$$f'(x) = \frac{e^x(x+1) - e^x \cdot 1}{(x+1)^2}$$

$$f'(0) = \frac{e^0(0+1) - e^0 \cdot 1}{(0+1)^2} = \frac{0}{1} = 0 \text{ slope}$$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = 0(x - 0)$$

$$\boxed{y = 1}$$